

RESEARCH ARTICLE



Access to healthcare: waiting time until the surgical procedure

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ABSTRACT

Background: Understanding the determinants that limit the population's access to surgical care in health services is highly relevant in order to provide data to support political interventions.

Objective: This study aimed to evaluate the time between diagnosis and elective surgery in adult patients with the longest waiting lists in Brazil; identify the determinants that interfere with access to the health service to perform surgery; and analyze the quality of life after the indicative diagnosis of surgical intervention.

Methods: A cross-sectional study was conducted with adult patients treated at three hospitals in the Southeast, North and South regions of Brazil, from October 2020 to October 2022. Data collection included socio-demographic data and assessment of quality of life using the WHOQOL-Bref instrument in the immediate postoperative period and one month after surgery.

Results: A total of 250 patients participated in the study, 55.6% patients from the Southeast, 20.4% patients from the North and 24% patients from the South, with a mean age of 51.86 (SD = 14.27) years and clinical history such as arterial hypertension ($p < 0.001$). The longest mean waiting time for surgery identified was 26.23 (SD = 17.62) months in the South region, with a significant difference between the evaluated institutions ($p = 0.02$). Differences were observed between the first place of care of the evaluated patients ($p < 0.001$). There was a difference in the perception of quality of life between the immediate and late postoperative periods ($p = 0.007$) and in the physical domain, with an increase in scores among older patients ($p = 0.004$) and previous clinical history ($p = 0.03$).

Conclusion: Access to the health system varies by region and does not meet the standards proposed by the Brazilian health system. In addition, it seems that longer waiting times for surgery more perceptibly affect the quality of life of older adults and those with other associated diseases.

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Introduction

Surgery is an essential part of healthcare and progress is being made towards universal health coverage, but population access is uneven across countries. About five million people around the world do not have access to safe surgery and anesthesia, with the proportion in low-income countries estimated at 97% and in high-income countries a proportion of 14.9%, considering the dimensions of opportunity, surgical capacity related to workforce and infrastructure, safety and accessibility [1,2].

The surgical system is involved in the entire spectrum of the disease from prevention to diagnosis, treatment and palliative care, thus requiring human resources including clinicians, surgeons, anesthesiologists, nursing staff and professionals from radiology, pathology and laboratory services. In addition, materials and equipment, preoperative assessment of surgical risk, adequate surgical facilities and blood supplies are required [3].

The lack of access to surgical treatment, especially in low- and middle-income countries, leads to a growing

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increase in among the lower income population, resulting in important losses for the economy and gross domestic product (GDP) of countries. Therefore, it is essential to track the prevalence of surgical conditions and treatment in local, national and international monitoring systems [4].

As a result, the World Health Organization (WHO) proposed a resolution for countries to develop national health plans with priority for surgical and anesthetic care in 2015, highlighting the importance of service provision, quality, training, workforce, infrastructure and information collection for monitoring and evaluating actions [5].

Health in Brazil is understood as a fundamental human right by law, and it is up to the State to provide the indispensable conditions for its full exercise [6]. In this context, access to health services must be ensured to every citizen among the principles of the Brazilian Unified Health System (*Sistema Único de Saúde - SUS*), regardless of gender, race, occupation or social conditions, in addition to equal rights to services [6].

However, Brazil has great inequalities in access and healthcare in different regions of the country, and the unequal distribution of health is not a natural phenomenon, but the result of a combination of inefficient health policies and programs and unfair economic arrangements [7]. In addition, studies point out that lower education, income, and place of residence stand out among the social determinants for difference in access to surgeries [8,9].

The difference in the population's access to surgical services in Brazil is more relevant in the North region, with a lower density of professionals to assist patients who seek care at health units and a lower volume of surgeries performed [10]. Added to this, analysis of data from the Unified Health System (*DATASUS*) in the period from 2008 to 2016 pointed to an increasing and significant evolution of surgical mortality, with an overall rate of 1.63% among the different specializations, and variation from 0.02% to 11.87% between vision surgeries and thoracic surgeries, respectively. However, surgical hospitalization rates did not significantly increase and the volume of surgical procedures remained constant [11].

Therefore, understanding the determinants that limit the population's access to surgical care in health services is highly relevant in order to provide data to support political-economic interventions that can progressively reduce the prolonged waiting time for a procedure and the different appointments between health units.

Thus, this study was conducted in Brazil to: evaluate the time between the health diagnosis and the performance of elective cholecystectomy, varicose vein and

hernia repair surgery in adult patients; identify the determinants that interfere with access to the health service to perform elective surgery; and analyze the quality of life after the indicative diagnosis of elective surgical intervention.

Methods

A quantitative, cross-sectional study was conducted in three Brazilian university public hospitals in the North (Belém-PA), South (Porto Alegre-RS) and Southeast (São Paulo-SP) regions of Brazil, from October 2020 to October 2022.

The North (Belém-PA) and South (Porto Alegre-RS) hospitals are universities hospitals responsible for a significant part of the surgeries performed by the public health system in this municipalities, considering reference to medium and high complex level of care. The Southeast (São Paulo-SP) hospital is an institution under administration of state university, responsible by more than 400 thousand patients, who live in west region of the São Paulo city or work at the university.

The study was approved by the Ethics Committee under number 4.469.319. The study participants expressed their agreement to participate in the investigation by signing the Informed Consent Form.

Sample size, inclusion and exclusion criteria

Study participants were adults undergoing elective surgeries performed by the Brazilian public health system. The sample size was estimated from a previous study [12] that evaluated the time between symptoms and initiation of treatment, which allowed an estimate of a standard deviation of 13.73 months, considering a margin of error of two months and a confidence level of 95%. In this way, the study required the observation of 181 patients to obtain the mean time for treatment.

Patients aged 18 years or older who underwent elective surgery performed by the Brazilian public health system for cholecystectomy, varicose veins and hernia repair were included. These surgeries were chosen because Brazilian national data showed that the Unified Health System (*SUS*) waiting list for elective surgery reached more than 904 thousand procedures in 2017, which can be represented by almost one million of patients waiting to a surgery in public system. Among the surgeries, emphasis is mainly placed on facetectomy, cholecystectomy, varicose veins, hernia repair, tonsil and adenoid surgery [13].

In this study, elective surgeries were defined by procedures that are scheduled previously, which physician evaluation do not require an emergency procedure.

Cholecystectomy is defined as a gallbladder extraction, which can be performed open or videolaparoscopy interventions [14]. A hernia is the egress of an organ through the anatomical structure or muscular structure that conventionally confines it. This condition principally manifests in the abdominal enclosure, as the intestine forcefully passes through a vulnerable region in the wall, which can require a hernia repair procedure [15]. While varicose veins are enlarged and twisted veins in the leg, which can be removed to treat the damaged veins, complications, or correct a problem that is causing the varicose veins [16].

Patients who had some cognitive limitations or mental health problems to communicate or understand were excluded, considering the difficult of these patients to evaluate the quality of their lives and inabilities to evaluate the time related to previous diagnose or primary assistance.

Study procedures

Patients who met eligibility criteria were identified from the schedules of the selected procedures in the participating hospital institutions. Eligible patients were approached and invited to participate in the study. Data collection was started after clarifications and signing of the Informed Consent Form.

Data were obtained through interviews with the patient during the immediate postoperative period using the RedCap software program (Research Electronic Data Capture), performed by the researcher and/or collectors trained for the study. One month after the surgical intervention (late postoperative period), the research team contacted the patient by telephone and repeated data collection with the World Health Organization Quality of Life instrument (WHOQOL-Bref) and questioned him or her about the occurrence of surgical complications. The data such as income, civil status, education, occupation, place of first care, referrals institutions, and WHOQOL-Bref answers was self-reported, and clinical-surgical aspects were reviewed by collectors in medical charts of the patients.

Measurement

Data collection included measures of socio-demographic characteristics, clinical and surgical variables, time between seeking treatment, diagnosis and surgery and quality of life considering the period between the time of diagnosis and surgery.

Socio-demographic characteristics, clinical and surgical variables, time between seeking treatment, diagnosis and surgery were measured with a study-specific instrument.

Quality of life (defined as their individual perception of their position in life in the context of the culture and value system in which they live, in relation to their goals, expectations, standards and concerns) was evaluated by the WHOQOL-Bref instrument [17], considering the period between the time of diagnosis and surgery.

The WHOQOL-Bref consists of 26 questions, two of which are general quality of life questions and the other 24 encompass the physical, psychological, social relations and environment domains [17,18].

Data analysis

Data were analyzed using descriptive statistics (absolute and relative frequencies), central tendency measures (mean, median and standard deviation), inferential analysis using Pearson's chi-squared statistical tests, one-way ANOVA, simulated Pearson's chi-squared test, and the Kruskal-Wallis test.

A mixed-effects linear regression model was applied to assess the association of postoperative time on patients' perception of quality of life considering the interaction of immediate and late postoperative time with the operating hospital, time between diagnosis and surgery, age, canceled surgery, and medical history.

All the variables mentioned were entered as a fixed effect and the subjects were entered as a random effect. This implies a longitudinal model with a compound symmetry covariance structure defined by:

1. Physical~Time * Hospital * (Time between diagnosis and surgery+Age+Surgery canceled+Comorbidities) + (1 | ID)
2. Psychological~Time * Hospital * (Time between diagnosis and surgery+Age+Surgery canceled+Comorbidities) + (1 | ID)

where the 1 in front means we have random intercept and the | ID indicates that there is a random intercept for each subject.

Results

A total of 250 patients were included in the study, 139 (55.6%) in the Southeast region, 60 (24%) individuals in the South region and 51 (20.4%) individuals in the North region. One hundred and fifty-four (61.6%) of the participants were female, 126 (50.4%) were married and had a mean age of 51.86 (SD = 14.27) years (Table 1). The mean ages of patients from the Southeast, North and South was 48.40 (SD = 14.43), 53.91 (SD = 14.45), 58.15 (SD = 11.10), respectively, indicating a statistically significant (< 0.001) age difference between the regions.

Table 1. Distribution and comparison of patients included in the study according to gender, age, civil status, income, education, occupation, service location ($n=250$). São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

Variables	$n=250$	p
Gender, n (%)		0.64 [†]
Female	154 (61.60)	
Male	96 (38.40)	
Age (years)		
Mean (SD)	51.86 (14.27)	
Median	52.50	< 0.001 [‡]
Minimum-Maximum	18.51–81.18	
Civil status, n (%)		
Married	126 (50.40)	
Single	57 (22.80)	
Stable union	29 (11.60)	< 0.001 [§]
Widowed	20 (8)	
Divorced	15 (6)	
Separated	3 (1.20)	
Income, n (%)		
No income	1 (0.40)	
< 1 minimum monthly salary	7 (2.80)	
1–3 minimum monthly salary	174 (69.60)	< 0.001 [§]
4–6 minimum monthly salary	57 (22.80)	
Don't know how to respond	11 (4.40)	
Education, n (%)		
Did not attend school	7 (2.80)	
Incomplete elementary	59 (23.60)	
Completed elementary	28 (11.20)	
Incomplete high school	27 (10.80)	0.18 [†]
Completed high school	83 (33.20)	
Incomplete higher education	13 (5.20)	
Graduated	33 (13.20)	
Occupation, n (%)		
Retired	50 (20)	
Salary employee	79 (31.60)	
Autonomous/self-employed	61 (24.40)	
Unemployed	33 (13.20)	< 0.001 [§]
Does not work	2 (0.80)	
Others	25 (10)	
Service location, n (%)		
Southeast	139 (55.60)	–
South	60 (24)	
North	51 (20.40)	

Legend: SD: standard deviation. † Pearson's chi-squared test; ‡One-way ANOVA; §Simulated Pearson's chi-squared test.

Statistically significant differences were also observed in marital status, income and occupation (see Table 1). Also, the greater number of comorbidities among patients was seen in the South region (Table 2).

A diagnosis of systemic arterial hypertension ($p<0.001$) and American Society of Anesthesiologists (ASA) II patients ($p=0.01$) was the chief comorbidity observed. Most patients in each of the institutions sought the health service at the onset of symptoms ($p<0.001$) (Table 2).

Regarding the time between diagnosis and surgery, the longest mean time of 26.23 (SD= 17.62) months was observed in the South region, with a significant difference between the evaluated institutions ($p=0.02$) (Table 2).

There was a difference between the regions in the first place of health care services accessed ($p<0.001$). The first care accessed was at the emergency department among patients in the Southeast region (47;

33.81%) and at the Primary Care in the South (38; 63.33%) and North regions (18; 35.29%), with an average of two services sought by patients before surgical resolution of the health problem (Table 2).

Due to the lower number of referrals offered by the medical team, patients in the North region sought a greater number of health services to obtain an indicative diagnosis for surgery (Table 2). In addition, five (9.8%) patients reported access to the surgery hospital through the help of relatives and friends who worked in the health service.

Three (2.15%) patients in the Southeast region reported that they were able to schedule the surgery after contacting the hospital's ombudsman and social worker, and two (1.43%) reported help from relatives and friends to obtain a surgical vacancy at the hospital. Laparoscopic cholecystectomy (168; 67.20%) and conventional herniorrhaphy (29; 11.60%) predominated for the type of surgery.

The largest number of cancellations occurred in the Southeast region (66; 47.48%), followed by the South region (26; 43.33%), caused by Covid-19 pandemic (52; 78.78%), lack of anesthesiologist in the service (12; 18%) and the lack of hospital beds and/or intensive care unit (12; 18%) standing out as reasons.

An increase was observed in the total score of the WHOQOL-Bref instrument between immediate to later postoperative, being that in the Southeast the score diversified from 71.25 (SD = 12.78) to 73.40 (SD = 15.42), 63.82 (SD = 10.93) to 65.67 (SD = 9.72) in North hospital and 62.85 (SD = 8.57) to 69.66 (SD = 9.15) in South hospital.

When analyzed the WHOQOL-Bref domains, improvement of score were observed especially in the physical and psychological domains (Figures 1 and 2).

In the South and North hospitals, the longer the waiting time for surgery, the lower the score in the physical domain. Also, patients with comorbidities showed less quality of life (Figure 1).

The Figure 2 evidenced the age impacting the psychological domain of the patients according to the hospital.

In the analysis of the interaction between the immediate and later postoperative period and the place where the surgery was performed, the mixed effects model indicated a significant difference in the total score on the WHOQOL-Bref instrument between the evaluated hospitals ($p=0.007$), also with emphasis on the physical domain ($p=0.002$) (i.e. better scores were observed in the Southeast and North compared with patients attended in the South region) (Table 3).

In the analysis of the physical domain, the presence of comorbidities contributed to the increase in

Table 2. Comparison of patients in terms of comorbidities, ASA classification, time between seeking care and diagnosis, time between diagnosis and surgery, commuting time, place of first care, number of services sought by the patient, number of referrals ($n=250$). São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

Variables (n=250)	Hospitals			p
	Southeast	North	South	
Comorbidities, n (%)				
Diabetes mellitus	14 (10.07)	9 (17.65)	12 (20)	0.12 [†]
Systemic arterial hypertension	35 (25.18)	20 (39.22)	34 (56.67)	< 0.001 [†]
Dyslipidemia	11 (7.91)	5 (9.80)	5 (8.33)	0.91 [†]
ASA, n (%)				
I	32 (23.02)	16 (31.37)	4 (6.67)	0.01 [†]
II	103 (74.10)	32 (62.75)	51 (85)	
III	4 (2.88)	3 (5.88)	5 (8.33)	
Seeking care: first symptoms, n (%)				
Yes	108 (77.70)	30 (58.82)	56 (93.33)	< 0.001 [†]
No	31 (22.30)	21 (41.18)	4 (6.67)	
Time between seeking care and diagnosis (days)				
Mean (SD)	185 (724.74)	87.78 (205.25)	27.43 (91.44)	0.40 [‡]
Median	0	0	0	
Min-Max	0–6605	0–731	0–579	
Time between diagnosis and surgery (months)				
Mean (SD)	21.14 (16.85)	20.75 (31.15)	26.23 (17.62)	0.02 [‡]
Median	20.00	11.62	24.00	
Min-Max	0.12–75.32	0.22–207.82	1.21–93.70	
Commuting time: home-hospital (minutes)				
Mean (SD)	47.96 (64.58)	47.02 (66.13)	53.22 (53.40)	0.16 [§]
Median	30	30	40	
Min-Max	5–600	5–429	5–300	
Place of first care				
Primary care	47 (33.81)	18 (35.29)	38 (63.33)	< 0.001
OMC	10 (7.19)	2 (3.92)	–	
Hospital: emergency	47 (33.81)	5 (9.80)	10 (16.67)	
Hospital: clinic	28 (20.14)	8 (15.69)	10 (16.67)	
Private service	7 (5.04)	12 (23.53)	–	
Others	–	6 (11.76)	2 (3.33)	
Number of services sought				
Mean (SD)	2.40 (1.03)	2.63 (0.92)	1.90 (0.54)	<0.001 [†]
Median	2	3	2	
Min-Max	1–7	1–5	1–3	
Number of referrals				
Mean (SD)	1 (0.88)	0.75 (0.66)	0.92 (0.59)	0.21 [‡]
Median	1	1	1	
Min-Max	1–4	1–2	1–2	

Legend: SD: standard deviation; Min: minimum; Max: maximum; OMC: Outpatient medical care. † Pearson's chi-squared test; ‡ Kruskal-Wallis test; § One-way Anova.

postoperative scores, regardless of the hospital where the surgery was performed and the postoperative period, which indicates a better perception of quality of life after performing the intervention surgery (Table 4).

Considering the time between the diagnosis and the surgical intervention, an increase in the psychological domain score of the WHOQOL-Bref instrument was observed, influenced by age and the hospital where the patient underwent surgery (Table 4).

In assessing the social domain, both the patient's age ($p=0.04$) and the time between diagnosis and surgical intervention (0.004) had an interaction effect to increase the score in the social domain.

Lastly, no significant interaction effects were observed in the score change for the environment domain in the immediate and late postoperative period when evaluating the hospital, diagnosis time and clinical history.

Discussion

This study indicated differences between the regions of Brazil regarding the waiting time for surgical intervention and access to health services to obtain an early diagnosis and treatment of the health problems. The evaluation of these aspects is relevant since the time to perform surgery or surgical cancellation can directly impact the individual's health conditions and daily life, generating a worse perception of quality of life, increased pain, anxiety and postoperative complications [19].

While the majority of patients in the Southeast region initially sought care through emergency services, patients in the South and North regions accessed the healthcare system through the Basic Health Units of reference for their homes. However, individuals of a higher age group with a greater number of comorbidities were observed in the South region, and who waited a longer time for the

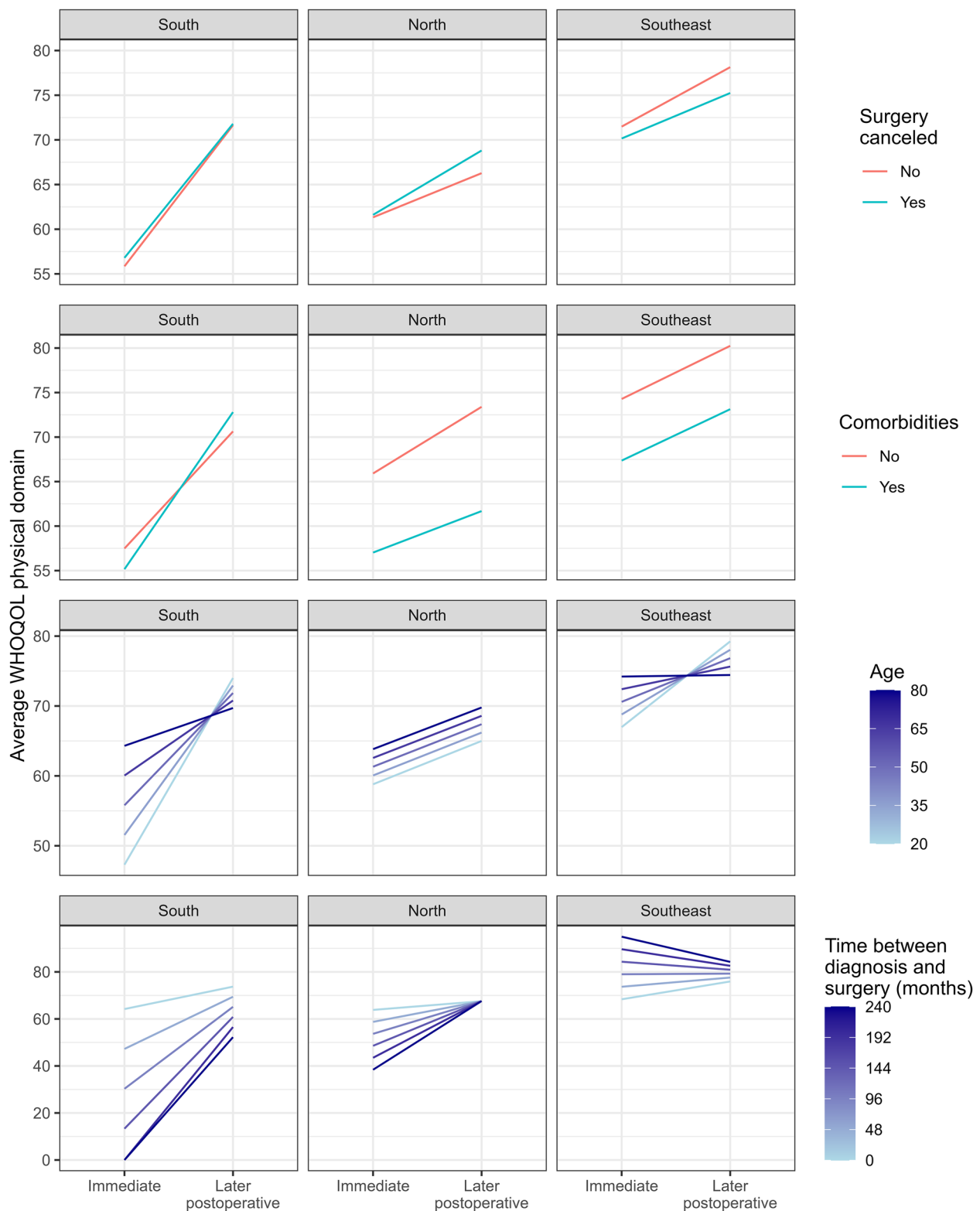


Figure 1. Physical domain score of WHOQOL-Bref between immediate and postoperative period, according to hospital and surgery canceled, comorbidities, age and time between diagnosis and surgery. São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

surgery to be performed, constituting an opposite result to that presented in a previous study which showed that patients with a greater number of previous health problems waited a shorter time to perform elective surgical

procedures [20]. Perhaps this process may be a result of the lack of a waiting list classification system in Brazil that defines priorities based on age and level of comorbidities.

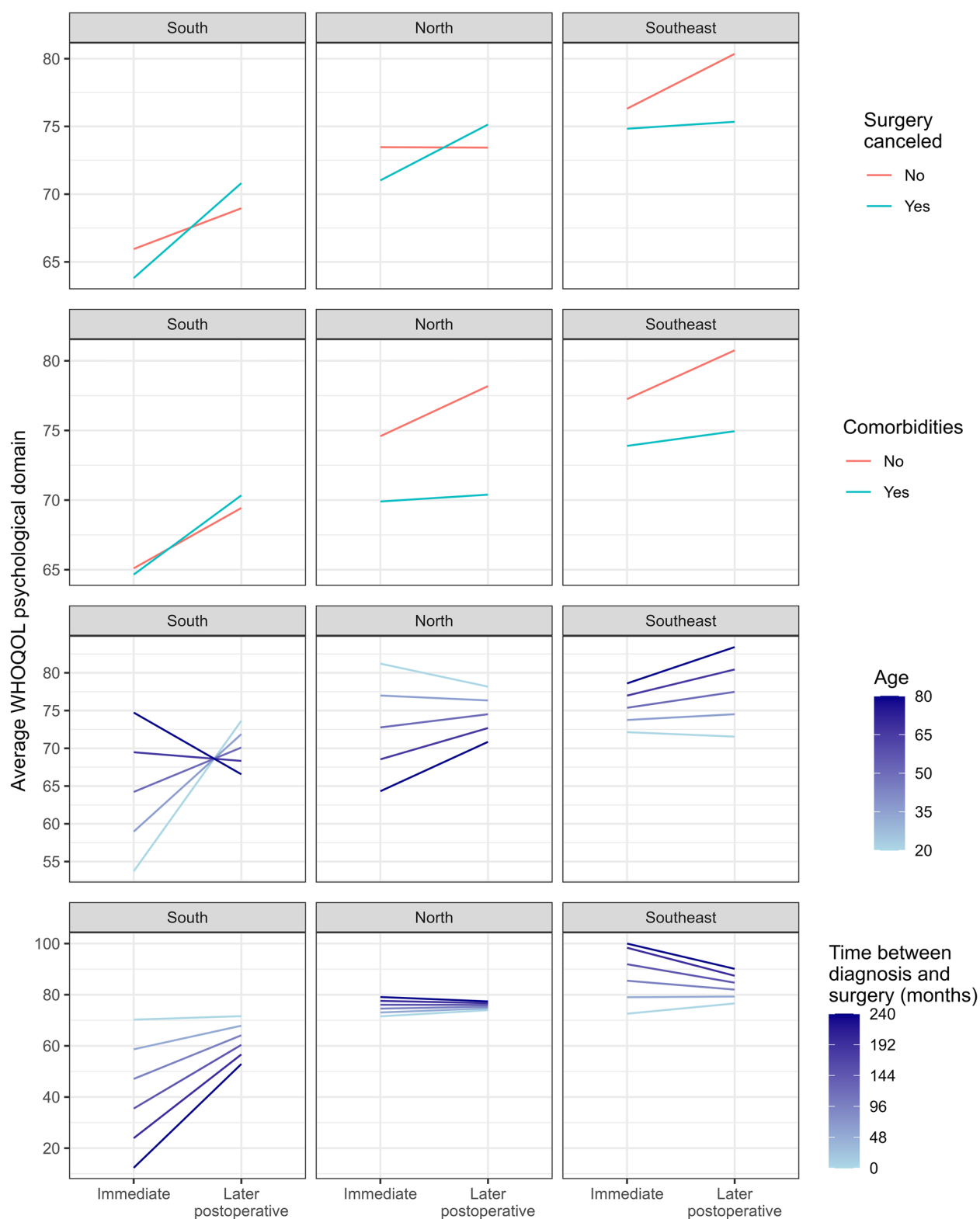


Figure 2. Psychological domain score of WHOQOL-Bref between immediate and postoperative period, according to hospital and surgery canceled, comorbidities, age and time between diagnosis and surgery. São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

On the other hand, patients from the North region received a smaller number of referrals in the places of care and waited less time to undergo the surgery

when compared to the South region. However, it should be noted that there was a higher frequency of assistance from relatives and friends to obtain a place

in the surgery hospital in the North region, meaning they sought alternative ways to gain referrals to access the Brazilian healthcare system. Therefore, patients in this region sought a greater number of health services to obtain a surgical diagnosis.

Thus, there is a weakness in the health system regarding the process of comprehensive care in the

sense of meeting the needs of the population. According to Ordinance No. 4,279, of December 30, 2010, the Healthcare Network (*Rede de Atenção a Saúde-RAS*) was created and defined as organizational arrangements of health actions and services, of different technological densities, which are integrated through technical, logistical and management support systems, seeking to ensure comprehensive care [21].

The objective of the RAS is to promote the systemic integration of actions and health services with the provision of continuous, integral, quality, responsible and humanized care, as well as to increase the performance of the healthcare system in terms of access, equity, clinical and sanitary efficacy; and economic efficiency. Thus, primary healthcare (PHC) in Brazil is structured as the first level of care and the gateway to the system, consisting of a multidisciplinary team that covers the entire population, integrating and coordinating care, and meeting their health needs [21].

However, the reference and counter-reference system (RCR) by the health services must work to operationalize the RAS, which refers to the mechanism for establishing communication between the different care levels in the health system. The objective of this system is continuity in the care offered to the user, which must be seen in its entirety through sharing health information at any level of the health system [22]. However, the study showed the patient's difficulty in obtaining continuity of care through referral via PHC, and the limitation of access to health services at the secondary and tertiary care levels, which could influence looking for more emergency services with

Table 3. Interaction between the operating hospital and the postoperative time on the total score and by domains of the WHOQOL-Bref instrument, estimated by the mixed-effects linear regression model ($n=250$). São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

	X ²	Df	p
Total score			
Intercept	7517.94	1	< 0.001
Time	25.84	1	< 0.001
Hospital	22.79	2	< 0.001
Time: hospital	9.91	2	0.007
Physical domain			
Intercept	4531.37	1	< 0.001
Time	52.78	1	< 0.001
Hospital	24.53	2	< 0.001
Time: hospital	1.25	2	0.002
Psychological domain			
Intercept	6501.80	1	< 0.001
Time	5.48	1	0.01
Hospital	17.29	2	< 0.001
Time: hospital	0.57	2	0.75
Social domain			
Intercept	4332.67	1	< 0.001
Time	6.77	1	0.009
Hospital	9.80	2	0.007
Time: hospital	5.49	2	0.06
Environment domain			
Intercept	5013.31	1	< 0.001
Time	0.21	1	0.64
Hospital	31.71	2	< 0.001
Time: hospital	2.23	2	0.32

Table 4. Interaction between postoperative time, hospital, age, time between diagnosis and surgery, medical history, canceled surgery on the physical domain score of the WHOQOL-Bref instrument, estimated by the mixed-effects linear regression model ($n=250$). São Paulo, Belém, Porto Alegre, Brazil. 2020–2022.

	Physical domain			Psychological domain		
	X ²	Df	p	X ²	Df	p
Intercept	162.73	1	< 0.001	241.58	1	< 0.001
Time	6.21	1	0.01	2.11	1	0.14
Hospital	1.00	2	0.60	2.91	2	0.23
Time between diagnosis and surgery	1.82	1	0.17	0.09	1	0.75
Age	0.59	1	0.44	0.07	1	0.78
Surgery canceled	0.001	1	0.98	0.35	1	0.55
Comorbidities	4.61	1	0.03	2.11	1	0.14
Time: Hospital	1.64	2	0.43	8.10	2	0.01
Time: Time between diagnosis and surgery	2.29	1	0.13	0.28	1	0.59
Time: age	2.55	1	0.11	0.89	1	0.34
Time: Surgery canceled	0.99	1	0.99	0.57	1	0.44
Time: Comorbidities	0.01	1	0.89	0.36	1	0.54
Hospital: Time between diagnosis and surgery	5.17	2	0.07	4.71	2	0.09
Hospital: Age	0.22	2	0.89	4.07	2	0.13
Hospital: Surgery canceled	0.56	2	0.75	0.71	2	0.70
Hospital: Comorbidities	1.83	2	0.40	0.95	2	0.62
Time: Hospital: Time between diagnosis and surgery	4.35	2	0.11	4.66	2	0.09
Time: Hospital: Age	1.15	2	0.56	11.21	2	0.004
Time: Hospital: Surgery canceled	0.24	2	0.88	4.90	2	0.08
Time: Hospital: Comorbidities	0.54	2	0.76	0.55	2	0.75

the aim to be assisted faster and receive an early diagnose.

Although global surgery is defined as an essential human right [23] and the Brazilian healthcare system advocates equal access to health services, waiting for surgical procedures in Brazil is still a reality in the national territory [13], and was expanded during the Covid-19 pandemic. The increased waiting period for surgical procedures during the pandemic increased the severity and consequently the postoperative morbidity and mortality, including complications, which contributed to executing procedures in the traditional modality, meaning open surgery, to the detriment of those performed by videolaparoscopy [24].

A previous analysis estimated a 40% reduction in surgical procedures in 2020 due to the restrictions imposed by the pandemic, with emphasis on a longer wait in the South region, with an estimated waiting time of 693 procedures per 100,000 inhabitants [25]. These data corroborate the findings identified in this study, which indicated the longest waiting time for surgical intervention among patients from Porto Alegre.

Structural aspects of the hospital health service also influenced the waiting time for the procedures in this study, such as the cancellation of surgeries due to lack of professionals (anesthesiologists), the lack of availability of hospital beds and limitations in access to surgical treatment due to the epidemiological picture of the Covid-19 pandemic. In this sense, the data show the weaknesses imposed by the reduced workforce in health services and the unequal distribution of professionals to adequately serve the population [26,27].

A previous study indicated an uneven distribution of surgeons, anesthesiologists and obstetricians in different regions of the country. Although Brazil has 34.7 medical professionals per 100,000 inhabitants, this density in the public health system is 23 professionals per 100,000 inhabitants, which in both cases is higher than the recommended minimum of 20 professionals; however, it contrasts the unequal distribution of professionals across Brazil [15]. This may be related to the lower supply of medical courses in the North region of Brazil, worse conditions and work overload, lower pay and infrastructure deficits in health services [28].

The analysis of quality of life after surgery indicated that patients with longer waiting times had lower scores on the WHOQOL-Bref instrument in the immediate postoperative period, which is a similar aspect observed in previous scientific literature that identified worse perception of quality of life among patients who have a longer waiting time for surgical intervention [19,29]. However, there was an increase in the total score of the WHOQOL-Bref instrument in the late

postoperative period, together with the physical and psychological domains, which represents a gain in terms of quality of life one month after performing minor surgical interventions, even after relatively long waiting periods having reduced the perception of quality of life of the analyzed subjects.

The waiting time for performing a surgery can generate or worsen anxiety and depression among patients with indication for procedures [20], which can intensify among patients who already have some previous comorbidity, in addition to increased pain, and limitations to perform daily activities and work [19].

The results of the study revealed irregular waiting periods between the regions of the country, failures in access to the healthcare system motivated by factors inherent to the system and/or motivated by reflections of the Covid-19 pandemic in the performance of elective surgeries. However, despite the negative impact on the patients' quality of life during the prolonged waiting period for the surgical resolution of the health problem, the surgeries led to an increase in the perception of quality of life within one month after the procedure.

Thus, guaranteeing quality surgical care capable of meeting the growing wait for this type of intervention should guide health practices and development of strategies and studies which seek to solve this serious public health problem. In addition, there is a need to develop a system for managing the waiting time for performing elective surgeries, with priority classification defined by the severity of clinical conditions and the need to perform the surgical intervention according to the risk level to the health of the individual patient.

Limitations

This study was conducted during Covid-19 pandemic, which could contribute to increase waiting time to surgery. However, the mean of waiting time to surgery was more than two years, which showed the weakness of public health system to provide health access to the population. Furthermore, all patients selected in this study were evaluated as no acute procedures, which classified them as elective patients and allowed portraying a real waiting time to the surgery at all hospitals.

Also, although the study was developed in three Brazilian hospitals, it portrayed the unequal access to the secondary and tertiary health care in different regions of the country.

Conclusion

The study identified a significant difference in waiting time for surgery between the different regions of the

country, especially in the South region, where patients waited more than two years after the diagnosis for the surgical procedure (26.23 months). In addition, it was observed that patients from the North region had a shorter waiting time to perform the surgery, but sought a greater number of health services to obtain a diagnosis and perform the surgical intervention, evidencing weaknesses in the functioning of the reference and counter-reference systems recommended by the SUS.

After performing the proposed elective surgery, the evaluated patients had a better perception of quality of life, with emphasis on the physical and psychological domains, associated with a greater increase in scores among older patients and those with a clinical medical history.

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Authors contribution

CSL, VBP, and NR conceived the idea for the study drawing from CSL pos-doctoral research. VBP and NR served as the pos doctoral supervisor, while AMM, DSTR served as an advisory group for the study. All authors provided input throughout the process. CSL, AYTA, KCA, FZM conducted the data collection and analysis, with oversight from VBP and NR, and wrote the first draft of the article with inputs from all authors. All authors edited the article and approved the final version.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Ethical approval and consent

The study was approved by the Ethics Committee of Nursing School- Sao Paulo University Ethics Committee, under number: 3.995.360, in April 26th, 2020, and Ethics Committee corresponding hospitals institutions. Also, the study adhered to the principles stated in the Declaration of Helsinki. Informed consent was obtained from all subjects involved in the study. All patients that agreed to participate in the study signed Free and Clarified Consent Term (FCCT) elaborated in accordance with Resolution 466/12 of the Brazilian National Health Council.

Paper context

Considering the territorial extension and unequal distribution of income, access to surgery in Brazil presents challenges related to the residence location, socio-demographic and economic issues,

and also the uneven availability of trained health professionals in the territory. The study showed the patient's difficulty in obtaining continuity of care through referral *via* primary health care, and the limitation of access to health services at the secondary and tertiary care levels.

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Data availability statement

The data will be shared upon request made to the corresponding author of the manuscript.

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